

Week 3 MPC Formulation: Building the Double Pendulum Cost and Constraints

1. The Baseline: Single Pendulum (Week 2)

In the single pendulum MPC formulation, the system was defined by a 2-dimensional state vector and a 1-dimensional control input:

- **State** ($x \in \mathbb{R}^2$): $x = [\theta, \dot{\theta}]^T$
- **Control** ($u \in \mathbb{R}^1$): $u = [\tau]$

The cost function penalized deviations from the upright state and excessive torque:

$$J_{single} = \sum_{k=0}^{N-1} \left(q_{\theta} \theta_k^2 + q_{\dot{\theta}} \dot{\theta}_k^2 + r_{\tau} \tau_k^2 \right) + \text{Terminal Cost} \quad (1)$$

2. Expanding to the Double Pendulum (Week 3)

To control the double pendulum, we build directly upon this foundation. The system gains a second degree of freedom, meaning we must track the position and velocity of the second link, and command a second motor.

The vectors strictly expand to accommodate the new link:

- **Expanded State** ($x \in \mathbb{R}^4$): $x = [\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2]^T$
- **Expanded Control** ($u \in \mathbb{R}^2$): $u = [\tau_1, \tau_2]^T$

3. The Double Pendulum Cost Function

Because the complex physical coupling (Coriolis, centrifugal, and inertial forces) is handled entirely by the nonlinear dynamics model, our MPC cost matrices Q and R can remain diagonal. We simply expand them to penalize the new states and controls independently.

The objective function remains a finite-horizon quadratic cost:

$$J(x, u) = \sum_{k=0}^{N-1} \left(x_k^T Q x_k + u_k^T R u_k \right) + x_N^T Q_{term} x_N \quad (2)$$

Where the matrices are built directly from our 1D logic:

- $Q = \text{diag}(q_{\theta_1}, q_{\dot{\theta}_1}, q_{\theta_2}, q_{\dot{\theta}_2})$
- $R = \text{diag}(r_{\tau_1}, r_{\tau_2})$
- $Q_{term} = 10 \times Q$ (Amplified to ensure local stability at the horizon's end).

Expanding this matrix multiplication yields the explicit formulation:

$$J = \sum_{k=0}^{N-1} \left(q_{\theta_1} \theta_{1,k}^2 + q_{\dot{\theta}_1} \dot{\theta}_{1,k}^2 + q_{\theta_2} \theta_{2,k}^2 + q_{\dot{\theta}_2} \dot{\theta}_{2,k}^2 + r_{\tau_1} \tau_{1,k}^2 + r_{\tau_2} \tau_{2,k}^2 \right) + \text{Terminal Cost} \quad (3)$$

4. Optimization Constraints

To ensure the solver outputs physically viable commands, we subject the objective function to constraints that mirror our single pendulum logic, scaled for the new dimensions:

A. Initial Condition Constraint

The solver must initialize at the true sensor readings:

$$x_0 = x_{sensor} \tag{4}$$

B. System Dynamics Constraint (Multiple-Shooting)

The predicted states must adhere to the physical laws of the 2-DOF pendulum. We enforce this using a 4th-order Runge-Kutta (RK4) integration of the provided Equations of Motion (EOM) over timestep $dt = 0.02s$:

$$x_{k+1} = f_{RK4}(x_k, u_k) \quad \text{for } k = 0, 1, \dots, N-1 \tag{5}$$

C. Actuator Saturation (Input Bounds)

Both physical motors possess a maximum torque limit ($\tau_{max} = 5.0 \text{ Nm}$). The solver is strictly bounded to prevent requesting impossible torques on either joint:

$$-\tau_{max} \leq \tau_{1,k} \leq \tau_{max} \tag{6}$$

$$-\tau_{max} \leq \tau_{2,k} \leq \tau_{max} \tag{7}$$